

13-3 The Various Measures of Cost

Our analysis of Chloe’s Cookie Factory showed how a firm’s total cost reflects its production function. From data on a firm’s total cost, we can derive several related measures of cost, which we will use to analyze production and pricing decisions in future chapters. To see how these related measures are derived, we consider the example in [Table 2](#). This table presents cost data on Chloe’s neighbor—Caleb’s Coffee Shop.

Table 2

The Various Measures of Cost: Caleb’s Coffee Shop

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Output (cups of coffee per hour)	Total Cost	Fixed Cost	Variable Cost	Average Fixed Cost	Average Variable Cost	Average Total Cost	Marginal Cost
0	\$3.00	\$3.00	\$0.00	—	—	—	
							\$0.30
1	3.30	3.00	0.30	\$3.00	\$0.30	\$3.30	
							0.50
2	3.80	3.00	0.80	1.50	0.40	1.90	
							0.70
3	4.50	3.00	1.50	1.00	0.50	1.50	
							0.90

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Output (cups of coffee per hour)	Total Cost	Fixed Cost	Variable Cost	Average Fixed Cost	Average Variable Cost	Average Total Cost	Marginal Cost
4	5.40	3.00	2.40	0.75	0.60	1.35	
							1.10
5	6.50	3.00	3.50	0.60	0.70	1.30	
							1.30
6	7.80	3.00	4.80	0.50	0.80	1.30	
							1.50
7	9.30	3.00	6.30	0.43	0.90	1.33	
							1.70
8	11.00	3.00	8.00	0.38	1.00	1.38	
							1.90
9	12.90	3.00	9.90	0.33	1.10	1.43	
							2.10
10	15.00	3.00	12.00	0.30	1.20	1.50	

Column (1) in the table shows the number of cups of coffee that Caleb might produce, ranging from 0 to 10 cups per hour. Column (2) shows Caleb's total cost of producing coffee. [Figure 3](#) plots Caleb's total-cost curve. The quantity of coffee [from column (1)] is on the

horizontal axis, and total cost [from column (2)] is on the vertical axis. Caleb's total-cost curve has a shape similar to Chloe's. In particular, it becomes steeper as the quantity produced rises, which (as we have discussed) reflects diminishing marginal product.

Figure 3

Caleb's Total-Cost Curve

Here the quantity of output produced (on the horizontal axis) is from column (1) in [Table 2](#), and the total cost (on the vertical axis) is from column (2). As in [Figure 2](#), the total-cost curve gets steeper as the quantity of output increases because of diminishing marginal product.

